

H1 - EEoS B prediction pipeline

This covers data sources, pipeline design, and methodological decisions for using the FishGlob data, combined with other datasources, to predict B at the haul level.

- It does not yet cover null / correlative model implementation.

Crucially: E is not available in FishGlob and must be derived externally. This document specifies how E is computed at the haul level, how predicted B is then generated using the micbru EEoS implementation, and records the decisions and assumptions involved.

High level flow:

- FishGlob haul_id → join to DATRAS on 7-field composite key → species length distributions (bins) per haul → FishBase LW parameters applied at bin level → haul-level E → micbru EEoS → B_pred → Residual = $\ln(B_{\text{obs}}) - \ln(B_{\text{pred}})$

1. State variables

The EEoS requires four state variables at the haul level. Three come directly from FishGlob; E must be derived.

Variable	Description	Source	Column
S	Species richness	FishGlob	distinct <code>accepted_name</code> per <code>haul_id</code>
N	Total abundance	FishGlob	<code>num</code> summed per haul
B_obs	Observed biomass (grams)	FishGlob	<code>wgt</code> summed per haul
E	Community metabolic rate, normalised	Derived	DATRAS HL + FishBase LW — Section 3

Panel: 182 ICES rectangles, 5,643 cell-years, Q1 1983–present. H1 evaluated at individual haul level. H2/H3 aggregated to rectangle with rectangle as random effect.

2. DATRAS data structure

NS-IBTS data retrieved via `icesDatras::getDATRAS()`. I tested the join between FishGlob and DATRAS on **2010 Q1**: `StatRec` = `stat_rec` in all cases.

HH — haul header

One row per haul. Provides the composite join key: Year, Quarter, Country, Ship, Gear, SweepLngt, GearEx, DoorType, StNo, HaulNo. FishGlob's `haul_id` encodes these same seven fields space-separated and is parsed to build the match.

HL — length frequency record

One row per haul × species × length bin. This is the primary input to E.

Field	What it contains	How it is used
<code>Valid_Aphia</code>	WoRMS species ID	Join key to FishGlob and FishBase lookup
<code>LngtCode</code>	Bin size and units: 1=1cm, .=1mm, 0=0.5cm	Determines unit conversion; all midpoints converted to cm
<code>LngtClass</code>	Lower bound of length bin	Midpoint = $\text{LngtClass} + 0.5$ (code 1) or $(\text{LngtClass} + 0.5)/10$ (codes ., 0)
<code>HLNoAtLngt</code>	Estimated number of fish at this bin in the full catch	Used as count weight in bin-level mass calculation — see note below
<code>SubFactor</code>	Subsampling expansion factor already applied to HLNoAtLngt	Diagnostic only; rows with SubFactor = -9 excluded

Note on HLNoAtLngt. When a crew cannot measure every fish, they measure a subsample and record an expansion factor (SubFactor). HLNoAtLngt already has this expansion applied — it represents the estimated count in the full catch at each length bin, not the raw measured count. Confirmed: in all 393 cases with $\text{SubFactor} > 1$, $\text{sum}(\text{HLNoAtLngt}) = \text{NoMeas} \times \text{SubFactor}$. $\text{SubFactor} > 1$ occurs in ~8.7% of rows; max observed = 1,888.

Potential bias: The expansion assumes the unmeasured fish have the same length distribution as the measured subsample. If subsampling was size-selective, HLNoAtLngt will misrepresent the true length distribution. This cannot be corrected from the available data.

Review point: Join validated on 2010 Q1 only. Verify match rates are consistent across all years during implementation. Flag haul-species combinations with high SubFactor and low NoMeas — the length distribution for those cases rests on very few measured fish.

3. E calculation pipeline

E is the total community metabolic rate, normalised so the smallest individual = 1:

$$E = \sum_s (n_s \cdot m_s^{(3/4)}) / m_{\min}^{(3/4)}$$

- n_s is abundance of species s (FishGlob `num`),
- m_s is mean individual mass in grams, and
- $m_{\min}$ is the mass of the smallest individual across all species in the haul.
- The normalisation is required by the micbru implementation — confirmed by inspection of `equation_of_state.py` where `lambda_2 = S / (E - N)` requires E and N in the same dimensionless units.

Step 1 — clean DATRAS HL

Pull NS-IBTS Q1 HL via `getDATRAS()` for all years 1983–present. Remove rows where `HLNoAtLngt` ≤ 0 , `LngtClass` ≤ 0 , or `SubFactor` = -9. Convert `LngtClass` to cm for `LngtCode` . and 0. Compute bin midpoint in cm for all rows.

Step 2 — build haul join key

Parse FishGlob `haul_id` by splitting on spaces to recover Year, Quarter, Country, Ship, Gear, StNo, HaulNo. Match to DATRAS HH on the same 7 fields.

Step 3 — length fallback hierarchy

For species present in a FishGlob haul but with no HL records, a representative length is assigned using the following fallback in order. A `length_source` flag records which level was used.

Level	Flag	Source
0	<code>HL_haul</code>	HL records exist for this haul-species — full length distribution available, proceed to Step 4

Level	Flag	Source
1	HL_rect_year	Mean of haul-level means for this species within the same rectangle and year
2	HL_species	Grand mean for this species across all Q1 DATRAS hauls 1983–present
3	FB_popchar	Mean length from <code>rfishbase::popchar()</code>
4	VB_approx	$L_{\infty} \times 0.67$ from <code>rfishbase::popgrowth()</code>
5	MISSING	No estimate obtainable — excluded from E; haul flagged if excluded species > 1% of haul N

Potential bias Levels 1–4 use progressively broader averages. If missing HL coverage is non-random — e.g. systematically higher for rare or bycatch species — the bias in E will be directional.

Review point Report fallback prevalence in methods. If more than 5% of haul-species combinations fall below Level 1, investigate the DATRAS HL join key across all years before proceeding.

Step 4 — bin-level mass and metabolic rate

For Level 0 species, the full length distribution is available. FishBase LW parameters are applied at each bin.

allometric equation $W = a \cdot L^b$: `m_k = a_s × midpoint_k ^ b_s` (mass in grams at bin k)

`epsilon_s = Σ_k ( HLNoAtLngt_k × m_k ^ 0.75 )`

`epsilon_s = n_s × (a_s × L_bar_s ^ b_s) ^ 0.75`

The `a_s` and `b_s` parameters come from the **FishBase LW lookup — described in Section 4**.

Step 5 — normalise and sum to haul-level E

Find the minimum metabolic rate across all species in the haul, then normalise and sum:

`epsilon_norm_s = epsilon_s / epsilon_min`

`E_haul = Σ_s ( epsilon_norm_s )`

For Level 0 species, `n_s` is embedded in the bin sum. For fallback species it enters explicitly. Record `prop_N_in_E` and `E_coverage_flag` (TRUE if `prop_N_in_E` < 0.99).

Review point : After normalisation, $\min(\text{epsilon_norm})$ must equal exactly 1 for every haul. Any haul where $E_{\text{haul}} < N$ indicates a normalisation or mass calculation error — exclude and investigate.

4. FishBase length–weight parameters

The LW relationship $W = a \cdot L^b$ converts a length measurement to a mass estimate. The parameters **a** and **b** are species-specific and are sourced from FishBase, which aggregates every published LW study it can find for each species.

- **b** is the slope of the relationship between length and mass: for most fish it is close to 3, reflecting the fact that mass scales roughly with the cube of length (a fish twice as long is roughly eight times heavier). **a** is a scaling coefficient that captures body shape and condition: a deep-bodied, well-fed fish has a higher **a** than a slender or lean one of the same length.

EG for herring this = 117 entries from different regions, seasons, and time periods, each an independently fitted regression from a different sample of fish.

A geographically matched parameter set is the ideal, but may not be possible for less studies species. FishBase entries include a **Locality** field with sufficient detail for North Sea filtering — for example, herring has entries explicitly labelled "North Sea / (Quarter 1) 1996–2008" and "North Sea / (January) 1996–2008". The proposed approach applies a four-level priority hierarchy, taking the median **a** and **b** from the highest level that yields at least three entries:

Level	Geographic scope	Season
1	North Sea and adjacent waters (ICES Sub-area IV, English Channel, Irish Sea, Skagerrak, Kattegat)	Q1-compatible (January, February, March, Quarter 1, or winter)
2	North Sea and adjacent waters	Any season
3	Northeast Atlantic broadly (Norwegian Sea, Celtic Sea, Bay of Biscay, wider UK and Atlantic European waters)	Any season

Level	Geographic scope	Season
4	Global — all entries with b in range 1–5	Any

This matters most for the species that dominate E. The top six species by N — herring (31.3%), sprat (24.0%), Norway pout (18.3%), whiting (9.4%), haddock (5.6%), dab (4.8%) — account for over 93% of total N. North Sea-specific FishBase entries exist for most of these.

The coverage analysis will report for each species: the geographic tier used, the resulting a and b versus the global median, and — most importantly — the N-weighted and B-weighted proportion of the panel resolved at each tier. A small number of species resolved at Level 1 may cover a large proportion of total N and E, which is the relevant metric for assessing the value of the refinement.

The `lw_source` flag will record both the taxonomic fallback level (direct / genus_mean / family_mean) and the geographic tier, providing a full audit trail.

Current coverage

Taxonomic coverage level	Species	N-weighted coverage
Direct FishBase parameters (TL-specific)	143	~99.5% of total N
Genus-level mean	23	Included in ~99.5%
Family-level mean	27	
Unresolved (Argentina spp. most significant)	6	~0.5% of total N

Review point — Fallback options for the 6 unresolved species have not been decided. Must be resolved before the pipeline is finalised. Argentina spp. is the only case with practical significance (~0.13% of total N).

Potential bias — Static FishBase parameters cannot capture temporal variation in body condition across a 40-year panel. The a parameter in particular varies with temperature, prey availability, and population age structure. This is an inherent limitation of the approach.

5. EEoS evaluation: micbru implementation

Once E is computed, S, N, E, and B_obs are assembled per haul and passed to `github.com/micbru/equation_of_state` (Python, called from R via `reticulate`). The `biomass(s)` function numerically solves for the METE Lagrange multipliers and integrates the metabolic rate distribution to return B_pred. The closed-form `biomass_approx(s)` will also be evaluated; `biomass()` is preferred as it does not rely on approximation conditions holding across the full range of S, N, E values in the panel. Approximately 14,000 calls for the full Q1 panel — runtime to be checked on a subset first.

The H1 residual:

```
residual = ln(B_obs) - ln(B_pred)
```

H2 and H3 test whether the absolute value of this residual is associated with fishing pressure.

Review point The `biomass()` function may not converge for all S, N, E combinations. Log hauls where constraints are not satisfied within 5% and examine whether they cluster by year, rectangle, or extreme state variable values. Exclude non-converging hauls from H1.

Review point The null model pipeline (permuted null ×999, ln(E) OLS comparator, unconstrained OLS) is specified but parked. Must be run before H2/H3 results are interpreted.

Potential bias B_obs includes all taxa in the FishGlob catch, including species with no FishBase LW parameters that do not contribute to E. If excluded species are systematically associated with fishing pressure or habitat type, the residual will carry structure unrelated to EEOs prediction quality.

6. Validation checks

Check	Pass criterion	Action if fail
Normalisation	$\min(\text{epsilon_norm}) = 1$ for every haul	Do not proceed
$E > N$	$E_{\text{haul}} > N$ for every haul	Investigate mass calculation

Check	Pass criterion	Action if fail
Toy example	Manual calculation matches pipeline output exactly	Do not proceed
Join match rate	Consistent across all years	Investigate DATRAS archive gaps
Fallback prevalence	>90% of haul-species at Level 0	Investigate join key specification
E coverage flag rate	<10% of hauls flagged	Investigate LW lookup gaps
EEoS convergence	Constraints satisfied within 5% for all hauls	Log and exclude non-converging hauls
Residual distribution	Approximately symmetric; no year or rectangle trend before fishing pressure regression	Systematic bias in E or B_obs